



ACIT 2620

Principles of Enterprise Networking

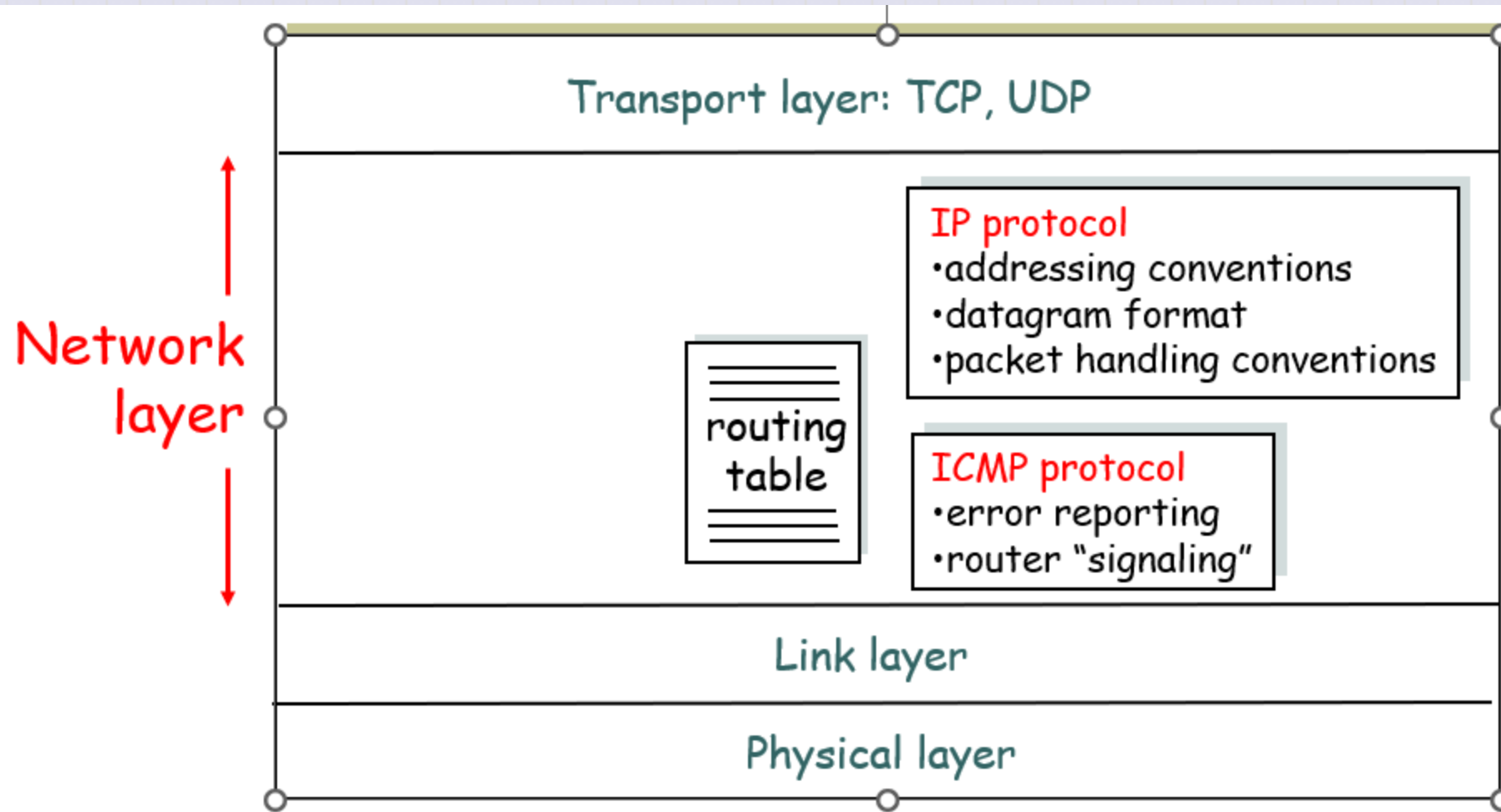
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Internet Protocol Version 4

Network Layer

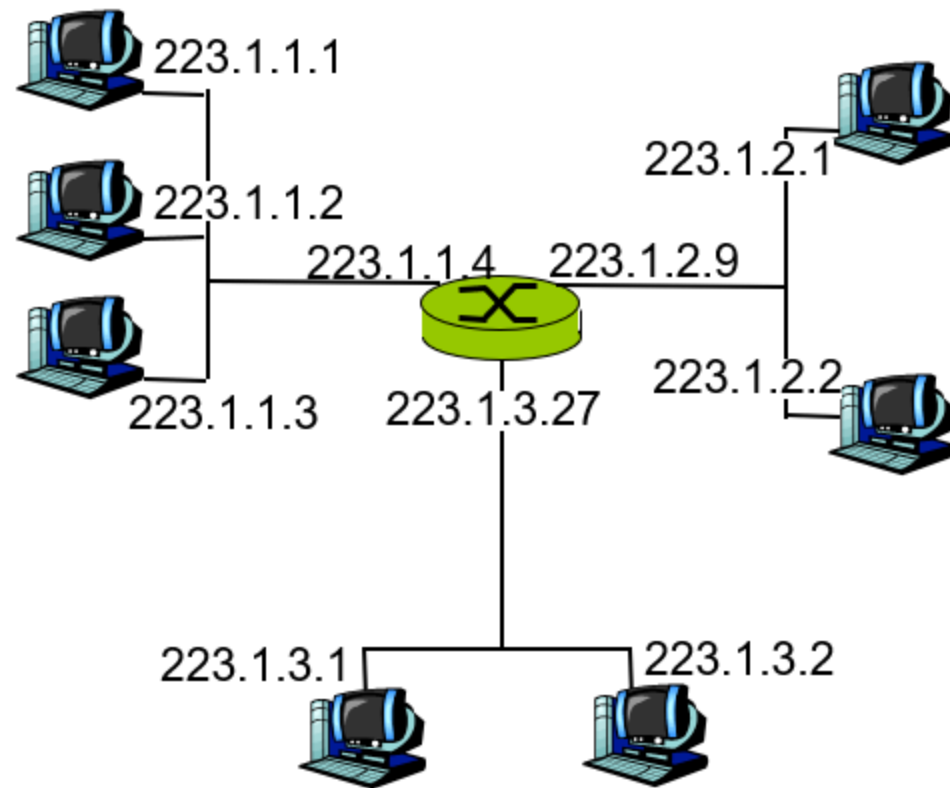
- **Goal:** move packets for source to destination
 - **Path Determination:**
the calculation of the route taken by packets -> routing
 - **Forwarding:**
The movement of a packet from one network to the next appropriate network

Network Layer functions



IP Addressing

- **IP address:** 32-bit identifier for host, router interface
- **interface:** connection between host, router and physical link
 - routers typically have multiple interfaces
 - host may have multiple interfaces
 - IP addresses associated with interface, not host, router



223.1.1.1 = $\underbrace{11011111}_{223} \underbrace{00000001}_1 \underbrace{00000001}_1 \underbrace{00000001}_1$

Components of an IP address

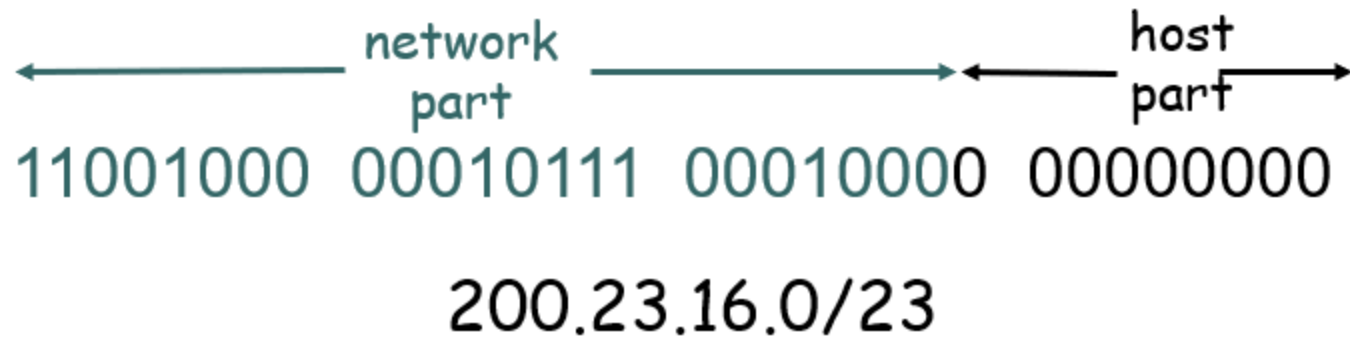
- network part (high order bits)
- host part (low order bits)

IP Network

- device interfaces with same network part of IP address
- can physically reach each other without intervening router

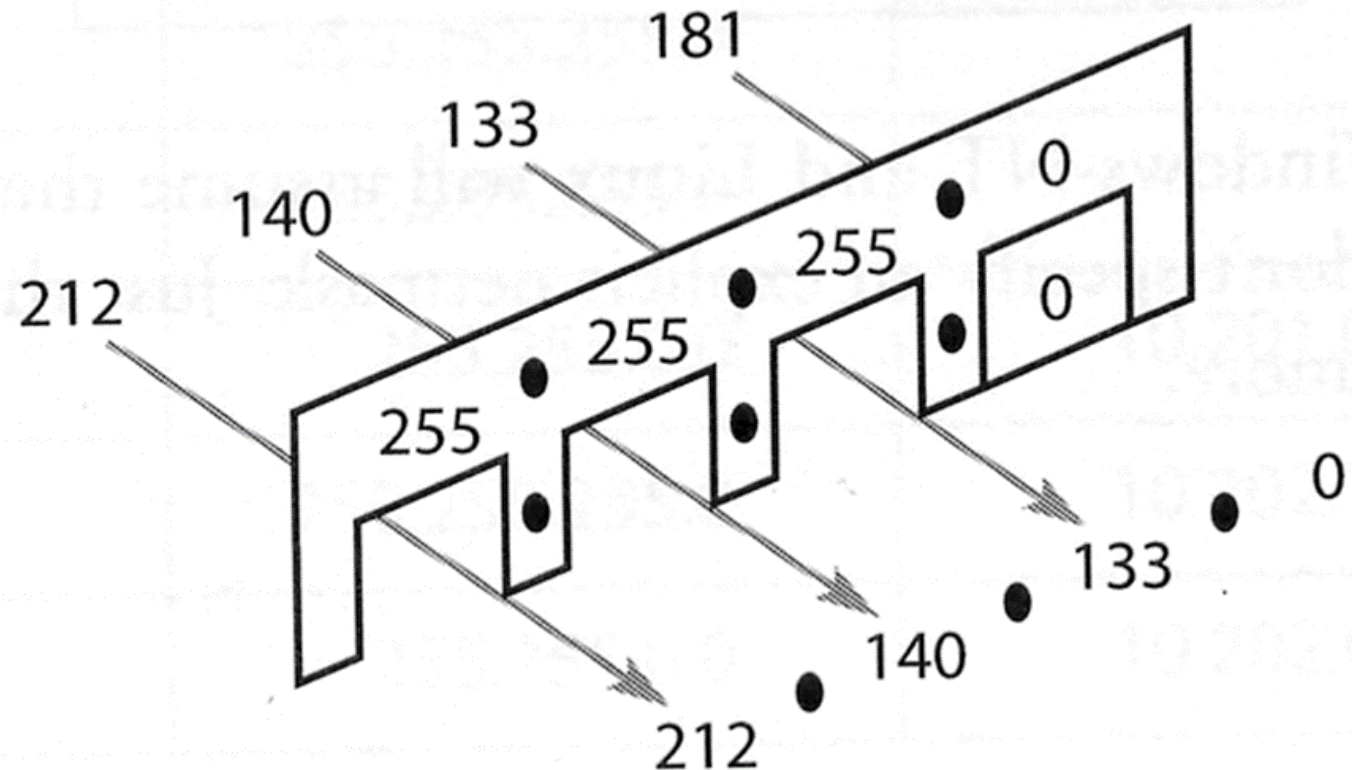
IP range notations

- **CIDR:** Classless Inter Domain Routing
 - network portion of address of arbitrary length
 - address format: **a . b . c . d / x** , where **x** is # of bits in network portion of address
 - Also written as address + subnet mask



- CIDR using Subnet Mask
 - Usually written in dotted decimal notation reminiscent of an IP address

Using mask 255.255.255.0



Special Addresses

- Private IP Addresses
 - 10.0.0.0/8
 - 172.16.0.0/12 (172.16 - 172.31)
 - 192.168.0.0/16

- Documentation IP Addresses
 - 192.0.2.0/24
 - 203.0.113.0/24
 - 198.51.100.0/24
- Self-Configured IP Addresses (often DHCP Failure)
 - 169.254.0.0/16

- Unknown Address
 - 0.0.0.0
- Loopback Address
 - 127.0.0.1 (actually, 127.0.0.0/8)
- Network Address (All host bits set to 0)
 - E.g: 192.168.1.0, Subnet Mask = 255.255.255.0

Broadcasting and Multicasting

- Limited Broadcast
 - 255.255.255.255
 - Transmitted only on local segment - not routed
- Network Broadcast Address
 - Network Address + All host bits set to one
 - Network Address = 192.168.1.x
 - Network Broadcast Address = 192.168.1.255

- Multicast Address
 - Lie within the 224.0.0.0 /4 network

IPv4 Header

